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Natural Language Processing

A1.5: ChatGPT

Do you think it can be useful to your learning (not for cheating but for improving your understanding of concepts in a course)? Why or why not?

Yes, I do think it can be useful to anyone's learning, as it can adapt to the needs of the person's requirements and level of speech. It can be used by kids to learn in school and for graduates who are pursuing higher academic feats such as a Master or a PHD. It is true that it is not that smart, but I feel like some people also underestimate it. If you use it properly, it can be a life saver. So far, I would estimate it has saved me more than 50h of workload ever since I signed up last November while my academic performance has not dropped. Specifically, it has helped me understand better what I do, e.g., in coding, because sometimes it gives me methods that I didn't even know of and explains to me why it is using them (see Figures 1 & 2). It also performs spelling checks for the reports I feed it and adds suggestions. It is like having a buddy making his best to help you with constructive criticism and it is only getting smarter with time. There are many concerns about it and almost all of them are justified, but I think the disruption of large language models will be a positive thing for society.

Regarding cheating and its limitations, it is great at summarizing texts and helping with the process of brainstorming for any task (see Figures 3 & 4). It can also help writing formally and turning ideas you might have as bullet points into a coherent text. Nevertheless, it does not do well in posing the ideas or deepening into a specific topic. In this context, we have more time to do the creative and meaningful part of the process, while it speeds up the more tedious or less appealing parts. Using it is a choice, you can still write the whole thing but, personally, it helps me a lot, as I doubt how to phrase some specific sentences and it just speeds up the process so much. For me, it feels like Wikipedia: at first, teachers discouraged its use but ended up using it as, with time, it became a reliable source of information with proper references even for the most niche mathematical topics. Wikipedia has become a useful tool and pupils are still punished from misusing it (e.g. limiting to copy & paste). I think chatGPT is a much more powerful version of it that is not limited to just providing information and, since it is already here and probably wont leave, it is our duty to learn how to get the most out of the tool.

Figures 1 & 2. Fragments of a conversation where I asked ChatGPT for help with some basic queries in Python.

hi, how do i display 2 functions in 1 plot in python?

To display two functions in one plot in Python, you can use the Matplotlib library. Here's an example of how to do it:

python Copy code

```python
import matplotlib.pyplot as plt
import numpy as np

# define two functions
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.cos(x)

# plot both functions on the same plot
plt.plot(x, y1, label='sin(x)')
plt.plot(x, y2, label='cos(x)')

# add a legend to the plot
plt.legend()

# display the plot
plt.show()
```

In this example, we first define two functions `y1` and `y2` using the `numpy` library. We then create a range of x values using the `linspace` function from `numpy`. We then plot both functions using the `plot` function from Matplotlib, specifying a label for each function. Finally, we add a legend to the plot using the `legend` function and display the plot using the `show` function.

how do i change the color of a heatmap

To change the color of a heatmap, you can use the `cmap` parameter of the `imshow` or `pcolormesh` functions in `matplotlib`. `cmap` stands for "colormap", which is a list of colors that the heatmap uses to represent different values in the data.

For example, you can change the color of a heatmap to a colormap named "YlGnBu" by adding the `cmap` parameter with the value "YlGnBu" to the `imshow` function:

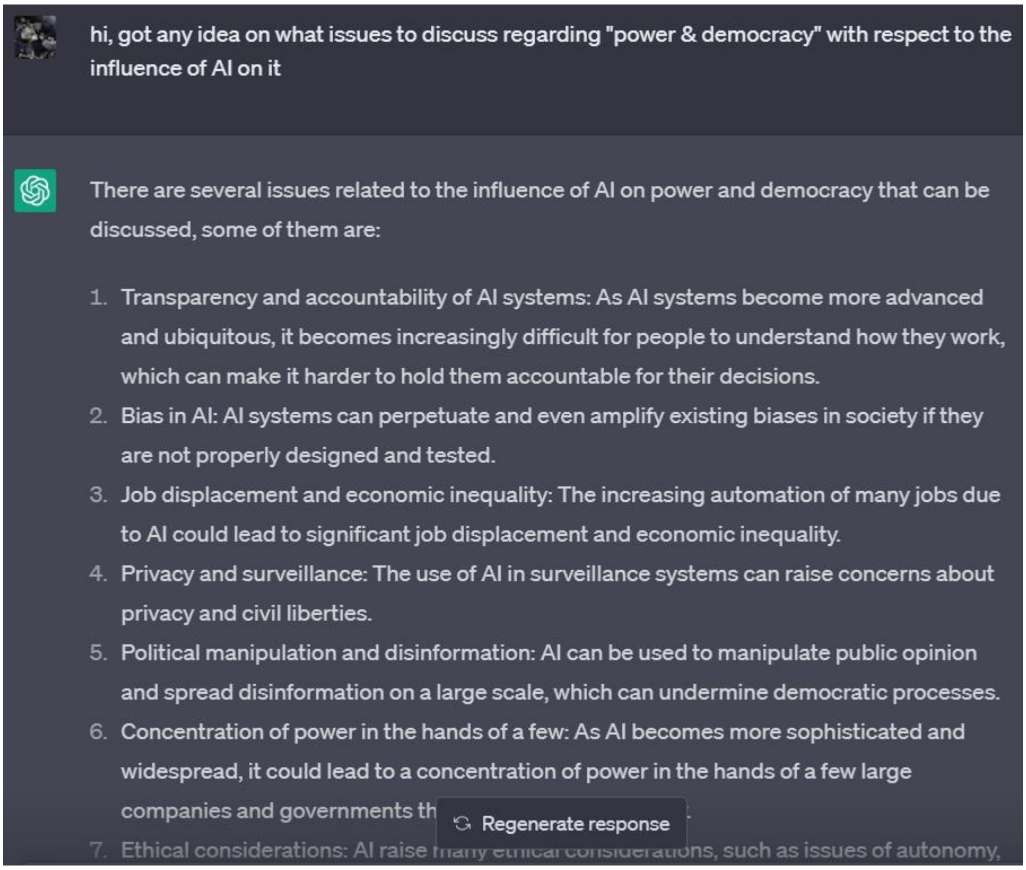
kotlin Copy code

```kotlin
import matplotlib.pyplot as plt
import numpy as np

data = np.random.rand(10, 10)
plt.imshow(data, cmap="YlGnBu")
plt.show()
```

There are many different colormaps available in `matplotlib`, so you can experiment with different ones to find the one that works best for your data. You can also create your own custom colormaps using the `ListedColormap` function.

Figures 3 & 4. Fragments of a conversation where I asked ChatGPT for help with a) brainstorming for issues regarding “Power & Democracy” and the role of AI on it and b) bullet points in favor of an opinion I wanted to express in an article.

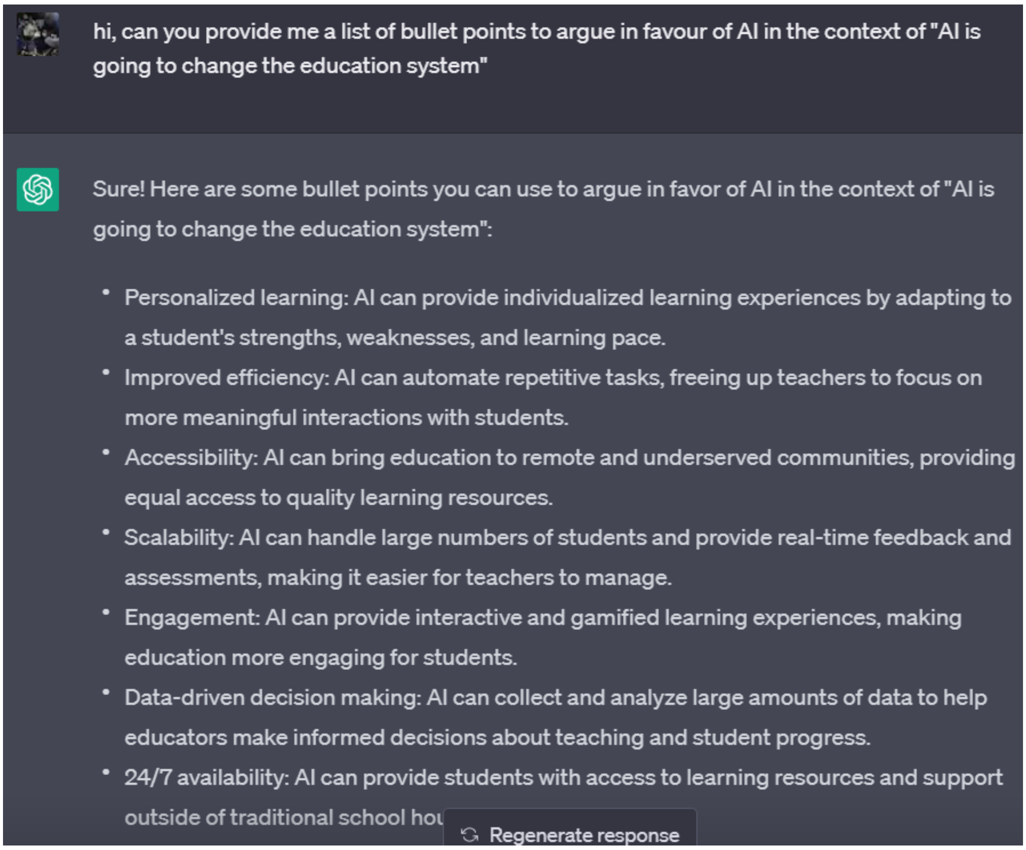


hi, got any idea on what issues to discuss regarding "power & democracy" with respect to the influence of AI on it

There are several issues related to the influence of AI on power and democracy that can be discussed, some of them are:

1. Transparency and accountability of AI systems: As AI systems become more advanced and ubiquitous, it becomes increasingly difficult for people to understand how they work, which can make it harder to hold them accountable for their decisions.
2. Bias in AI: AI systems can perpetuate and even amplify existing biases in society if they are not properly designed and tested.
3. Job displacement and economic inequality: The increasing automation of many jobs due to AI could lead to significant job displacement and economic inequality.
4. Privacy and surveillance: The use of AI in surveillance systems can raise concerns about privacy and civil liberties.
5. Political manipulation and disinformation: AI can be used to manipulate public opinion and spread disinformation on a large scale, which can undermine democratic processes.
6. Concentration of power in the hands of a few: As AI becomes more sophisticated and widespread, it could lead to a concentration of power in the hands of a few large companies and governments th
7. Ethical considerations: AI raise ns, such as issues of autonomy,

Regenerate response



hi, can you provide me a list of bullet points to argue in favour of AI in the context of "AI is going to change the education system"

Sure! Here are some bullet points you can use to argue in favor of AI in the context of "AI is going to change the education system":

- Personalized learning: AI can provide individualized learning experiences by adapting to a student's strengths, weaknesses, and learning pace.
- Improved efficiency: AI can automate repetitive tasks, freeing up teachers to focus on more meaningful interactions with students.
- Accessibility: AI can bring education to remote and underserved communities, providing equal access to quality learning resources.
- Scalability: AI can handle large numbers of students and provide real-time feedback and assessments, making it easier for teachers to manage.
- Engagement: AI can provide interactive and gamified learning experiences, making education more engaging for students.
- Data-driven decision making: AI can collect and analyze large amounts of data to help educators make informed decisions about teaching and student progress.
- 24/7 availability: AI can provide students with access to learning resources and support outside of traditional school hou

Regenerate response